Dual Space

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Linear Functional

A linear functional is a linear map from V to F.In other words,a linear functional is an element of $\mathcal{L}$(V,F)

Example 0.1. *Define* $\phi : \mathbf{R^3} \rightarrow \mathbf{R}$ *by* $\phi(x, y, z) = 4x-5y+2z. Then\ \phi is a linear functional on \mathbf{R^3}$

Example 0.2. *Fix* $(c_1, c_2, \ldots, c_n) \in \mathbf{F^n}.Define \phi : \mathbf{F^3} \rightarrow \mathbf{F}$ *by* $\phi(x_1, x_2, \ldots, x_n) = c_1x_1 + c_2x_2 + \cdots + c_nx_n$

Example 0.3. *Define* $\phi : \mathcal{P}(\mathbf{R})$ *by* $\phi(p) = 3p"(5)+7p(4)$ *Then* $\phi is a linear functional on \mathcal{P}(\mathbf{R})$

Definition 1. *The dual space of* V*,denoted by* V'*,is the vector space of all linear functional.In other words,* $V' = \mathcal{L}(V, \mathbf{F})$

Theorem 0.4. *Suppose* V *is finite dimensional.Then* V' *is also finite-dimensional and* $dimV = dimV'$

Proof. let $V' = \mathcal{L}(V, \mathbf{F})$,

$$\begin{aligned} Now, dim(V') &= dim(\mathcal{L}(V, \mathbf{F})) \\ &= dim(V).dim(\mathbf{F}) \\ &= dim(V) \qquad\qquad \because \text{dim(F)=1} \end{aligned}$$

$$\boxed{dim(V') = dim(V)}$$

hence, V' is finite dimensional and $dim(V') = dim(V)$ □

Definition 2. *if* $v_1, v_2, \ldots, v_n$ *is a basis of* V*,then the dual basis of* $v_1, v_2, \ldots, v_n$ *is the list of* $\phi_1, \phi_2, \ldots, \phi_n$ *of elements of* V'*,where each* ϕ_j *is the linqer functional on* V *such that*

$$\phi_j(v_k) = \begin{cases} 1, & \text{if } k = j, \\ 0, & \text{if } k \neq j \end{cases}$$

Theorem 0.5. *Suppose* $v_1, v_2, \ldots, v_n$ *is basis of* V *and* $\phi_1, \phi_2, \ldots, \phi_n$ *is the dual basis.Then*

$$v = \phi_1(v)v_1 + \cdots + \phi_n(v)v_n$$

for each $v \in V$

Proof. Suppose $v \in V$.Then their exist $c_1, c_2, \ldots, c_n \in \mathbf{F}$
such that

$$v = \sum_{i=1}^{n} c_i.v_i$$

consider,$\phi_j(v) = \phi_j(\sum_{i=1}^{n} c_i.v_i)$ where $j \in \{1, 2, \ldots, n\}$

$$\begin{aligned} &= \phi_j(c_1v_1) + \phi_j(c_2v_2) + \cdots + \phi_j(c_nv_n) \\ &= \phi_j(c_jv_j) \\ &= c_j\phi_j(v_j) \end{aligned}$$

$$\because \phi_j(v_j) = 1$$

$$\phi_j(v) = c_j$$

if $j = i$,then $\phi_j(v) = c_i$,
therefore, $\boxed{\text{v}=\sum_{i=1}^{n} \phi_i(v)vi}$

□